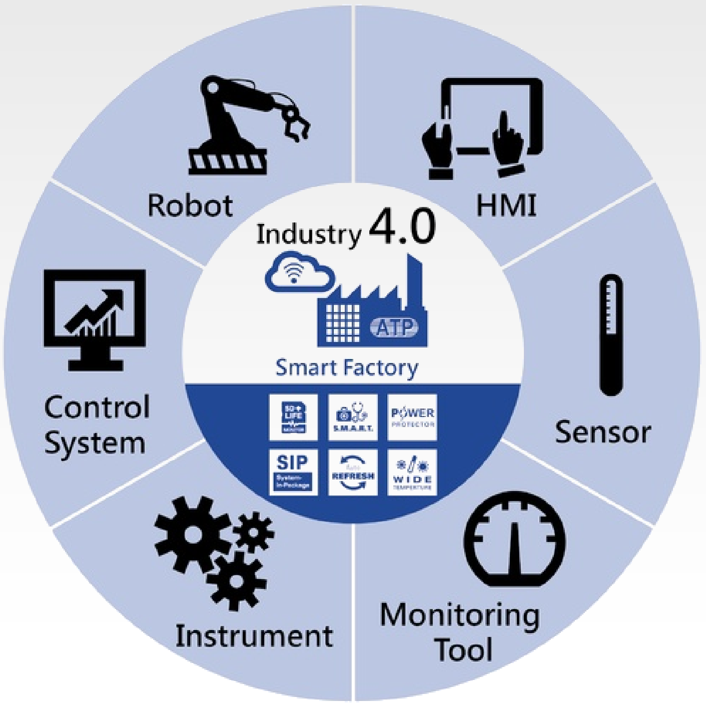
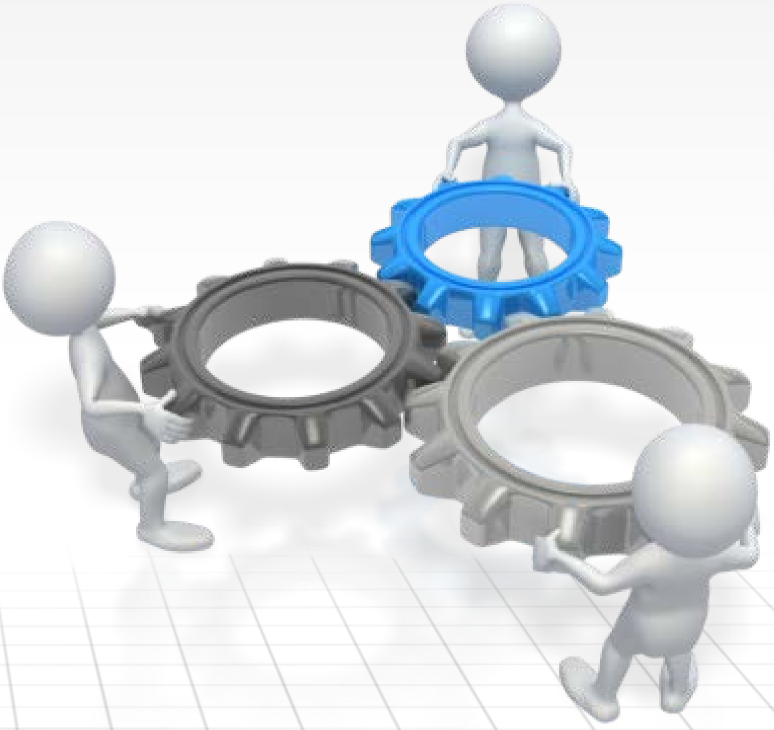


LECTURE 12

REVOLUTIONS AND EVOLUTIONS IN PRODUCTION

Contents



Objectives

1. Knowing the revolutions in production systems.
2. Understanding the technologies embedded in ultramodern manufacturing.
3. Knowing the characteristics, benefits and drawbacks of Industry 4.0.

Revolutions in production (1)

Quality of life
Engineering Sciences

1st

Mobility

2nd

μelectronics

3rd

ICT

4th

steam engine

GB

1782

Power generation
Mechanical automation

conveyor belt

US

1913

Industrialization

Computer, NC,PLC

US/EU

1954

Electronic Automation

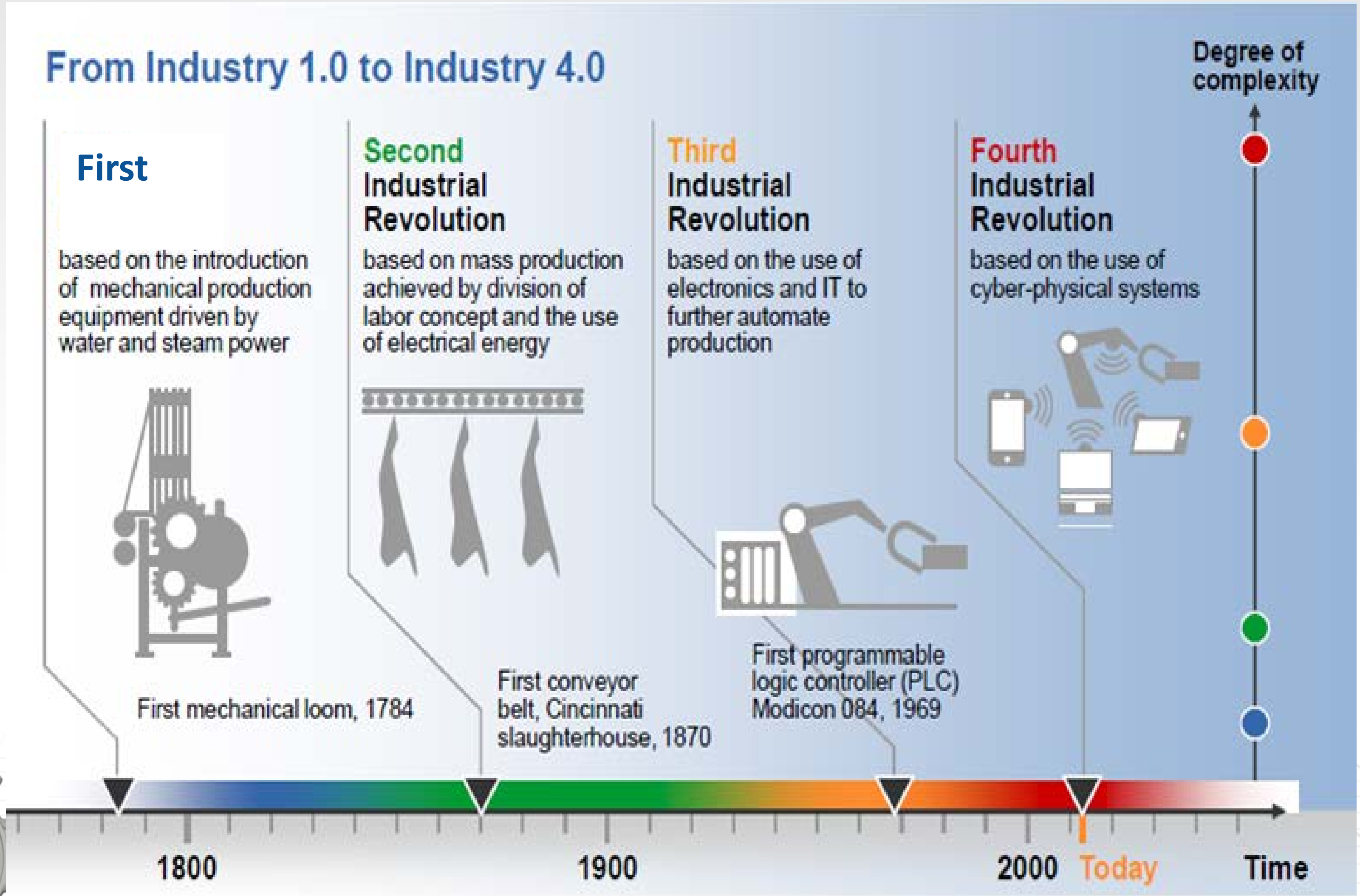
Cyber Physical Systems

EU

2015

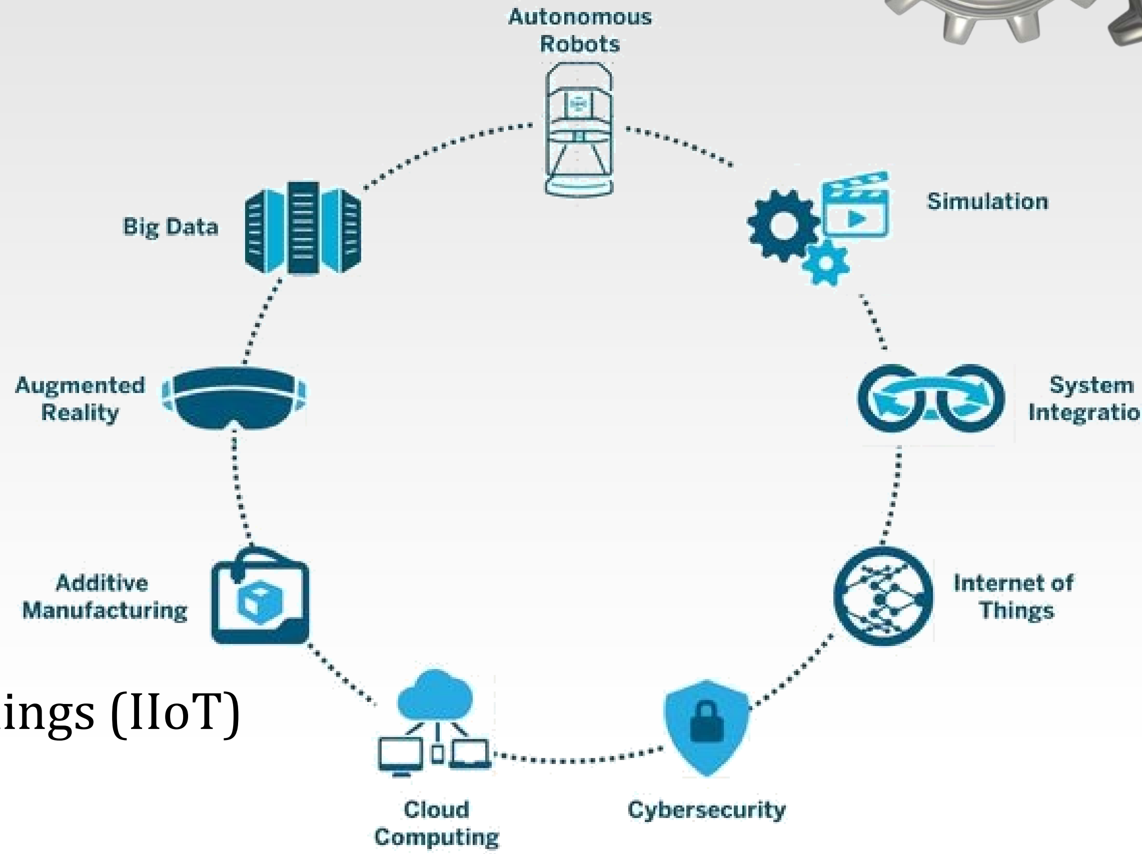
Smart Automation

Revolutions in production (2)



Modern technologies

- ❑ Digital factory
- ❑ Cloud computing
- ❑ Big data & analytics
- ❑ Cyber physical sistems
- ❑ Internet of things (IoT)
- ❑ Industrial internet of things (IIoT)
- ❑ Smart factory
- ❑ Industry 4.0



Self-assessment test

1. The ultra modern manufacturing involves the use of cyber-physical systems. **T** **F**

2. The ultra modern manufacturing does not involve the use of IoT. **T** **F**

3. A very large amount of data is acquired and processed in an ultra modern manufacturing system. **T** **F**

4. The Industry 4.0 has emerged with the first PLCs and computers. **T** **F**

5. The Cyber-physical systems are related to the third industrial revolution. **T** **F**

Digital factory

Digital Factory is the factory that uses information technology in all its activities.

- Activities where information technology is used:
  - design,
  - manufacturing,
  - assembly,
  - quality control,
  - planning,
  - management,
  - diagnosis and optimization.

- The ultimate goal is to:
  - maximize efficiency of production,
  - cut costs,
  - increase the quality of the products.

Cloud computing

Cloud computing is a set of distributed computing services, applications, access to information and data storage. The user does not need to know the physical location and configuration of systems providing these services.

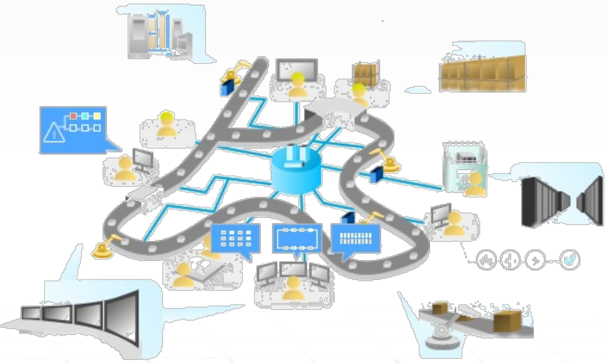
- **Advantages:**
  - Data syncronization between multiple devices is simplified.
  - The documents in cloud can be processed online, using web applications.
  - The computing speed and the storage capacity are high.
- **Disadvantages:**
  - It requires a fast and stable Internet connection.
  - Data security may be compromised.

Big data & analytics

Big data is a term for data sets that are so large or complex that traditional data processing application softwares are inadequate to deal with them.

- **Big data & analytics activities:** capture, storage, analysis, data curation, search, sharing, transfer, visualization, querying, updating and information privacy.

- **Big data** provides an infrastructure for managing the uncertainties (eg. uncertainties related to the performance or availability of the production equipment).

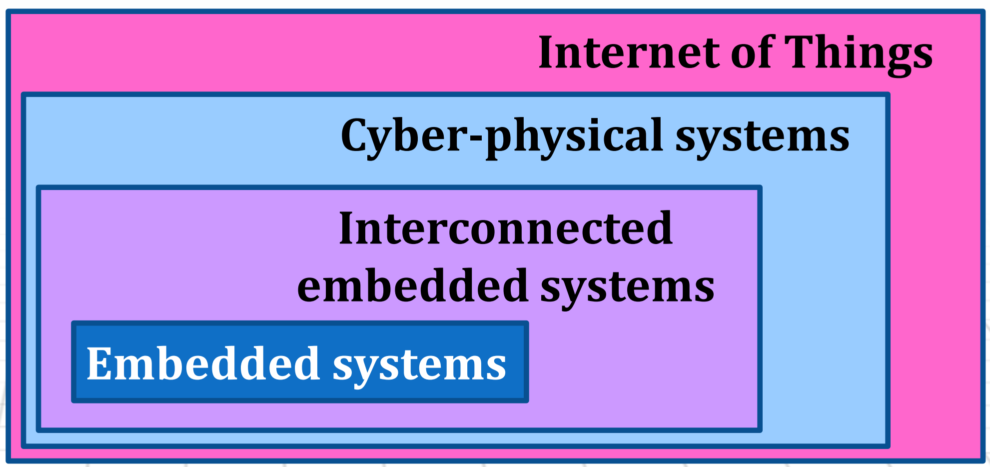


- In **predictive manufacturing** the operation of production systems is not affected by the failure of equipment. A large amount of data is collected from sensors and processed using big data & analytics technologies.

Cyber-physical systems

Cyber-physical systems bring together physical and virtual world, creating an interconnected world where smart objects communicate and interact.

- ❑ Cyber-physical systems are the next level in the evolution of embedded systems.
- ❑ Embedded systems together with Internet services create the cyber-physical systems.
- ❑ Cyber-physical systems provides the foundations for IoT.



Self-assessment test

1. The user of the cloud computing services has to know the physical location and configuration of equipment that provides these services. **T** **F**

2. Information technology is used in quality control. **T** **F**

3. The digital factory uses information technology in all its activities. **T** **F**

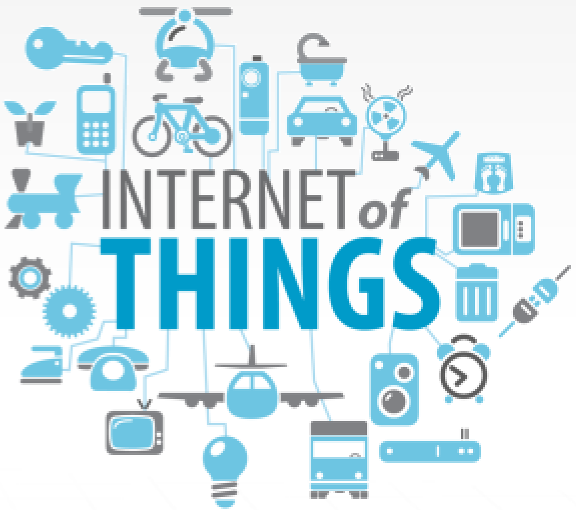
4. Predictive manufacturing systems require a large amount of data collected from sensors. **T** **F**

5. The embedded systems, encapsulate the cyber-physical systems. **T** **F**

Internet of things

Internet of Things (IoT) is a concept that defines a world where all products (machinery, appliances, lighting systems, mobile devices, etc.) are interconnected through the Internet.

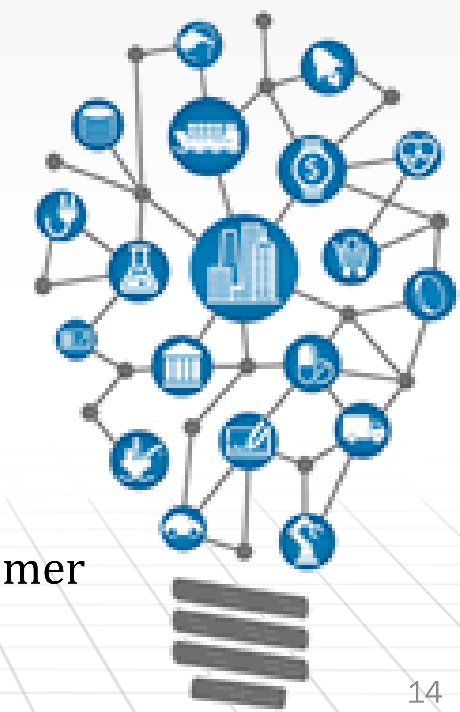
- IoT makes possible the detection and / or the remote control of the devices, via an existing network (wired or wireless).
- It is therefore possible a rapid integration of the computer-based equipment.
- Unfortunately, the devices connected to IoT are quite vulnerable to cyber attacks.



Industrial Internet of things

Industrial Internet of Things (IIoT) is a concept introduced by General Motors, which describes how Big Data analytics combined with IoT can provide opportunities for the industry.

- IIoT is the equivalent to the Industry 4.0 term, proposed by the Germans.
- Benefits of IIoT:
  - improves operational efficiency,
  - improves productivity,
  - creates new business opportunities,
  - reduces downtime,
  - maximizes asset utilization,
  - reduces the cost of assets,
  - increases worker safety,
  - improves innovation
  - make possible a better understanding of the customer needs.



Smart factory (1)

The smart manufacturing is the result of using the cyber-physical systems in production.

Smart phones

Smart Homes

Smart Cars

Market Pull

Smart Factories

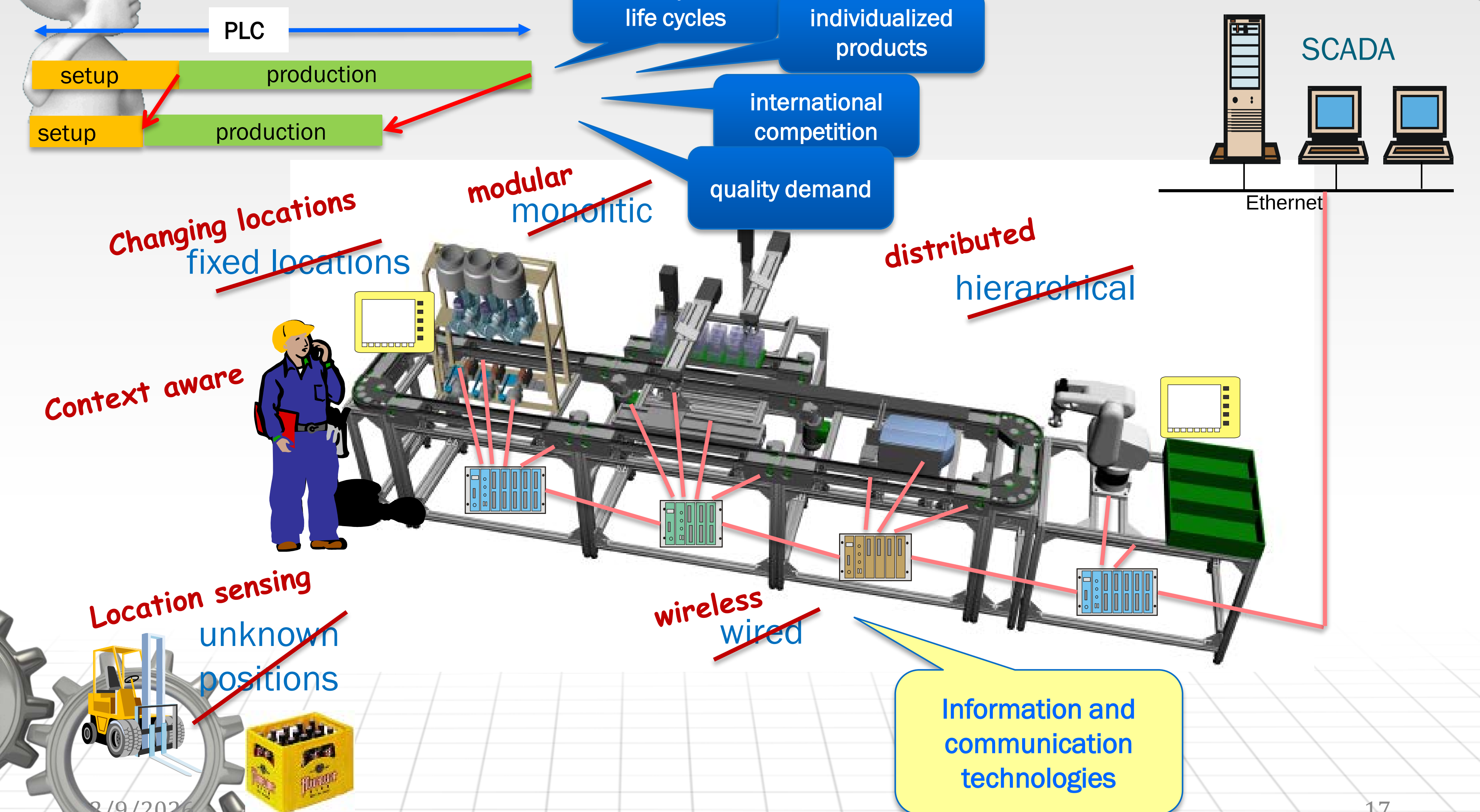
Technology Push

Smart factory (2)

- **Features** of Smart factories**:**
  - very high level of automation,
  - production systems with high flexibility thanks to interconnected networks of cyber-physical systems (CPS),
  - optimized production processes using global criteria.

- **Advantages** of Smart factories:
  - CPS-optimized production system;
  - optimized individual customer product manufacturing;
  - efficient allocation of resources;
  - the machine adapts to the human work cycle;
  - self-learning and self troubleshooting abilities.

Intelligent factory vs traditional factory (1)



Intelligent factory
vs
traditional factory
(2)

Challenges for industry are growing worldwide

Self-assessment test

1. In IoT, all the products are interconnected through the Internet. **T** **F**

2. IIoT has lower requirements than those specific IoT. **T** **F**

3. The cyber-physical systems are used in smart manufacturing. **T** **F**

4. The control system of a traditional factory is hierarchic. **T** **F**

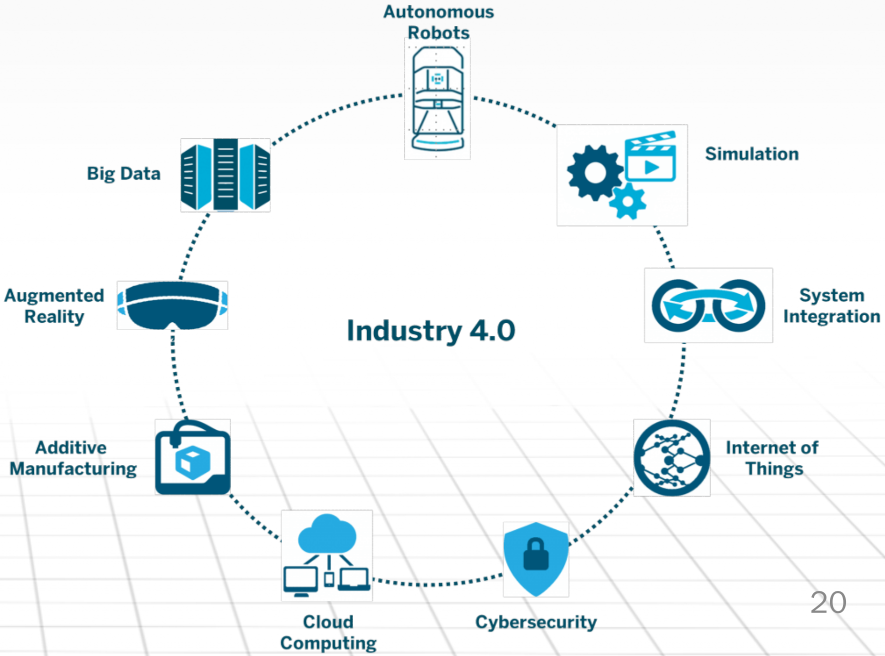
5. The decisions are context aware in smart factories. **T** **F**

Industry 4.0 (1)

Industry 4.0 is the current trend of automation and data exchange in manufacturing technologies. It includes cyber-physical systems, the Internet of things and cloud computing

- The term "Industrie 4.0" originates from a project in the high-tech strategy of the German government, which promotes the computerization of manufacturing.

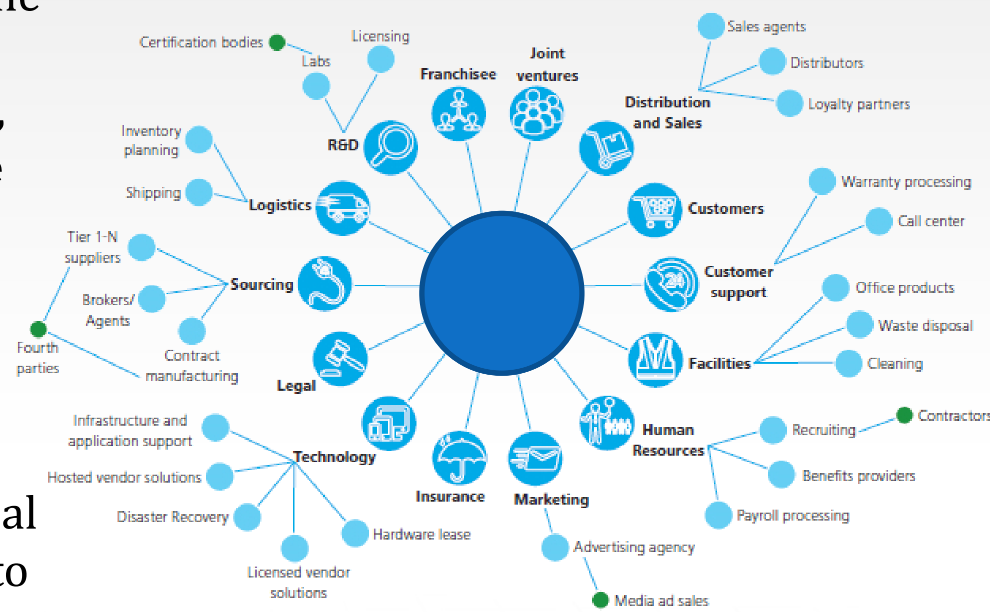
- **Terms** related to Industry 4.0:
  - industrial internet,
  - advanced manufacturing,
  - future factory,
  - cyber-physical system,
  - smart factory.



Industry 4.0 (2)

- *Industry 4.0* creates smart manufacturing.

- In the modular smart factory, the cyber-physical systems:
  - monitor physical processes,
  - creates a virtual copy of the real world,
  - take decisions in a decentralized way,
  - using IoT technologies communicate in real-time,
  - provide internal and external services to all participants to the production chain (design, production, sales, use and recycling).



Industry 4.0 (3)

- The main features of the Industry 4.0:
  - Interoperability,
  - Virtualization,
  - Decentralization,
  - Real-time capability,
  - Service orientation,
  - Modularity.

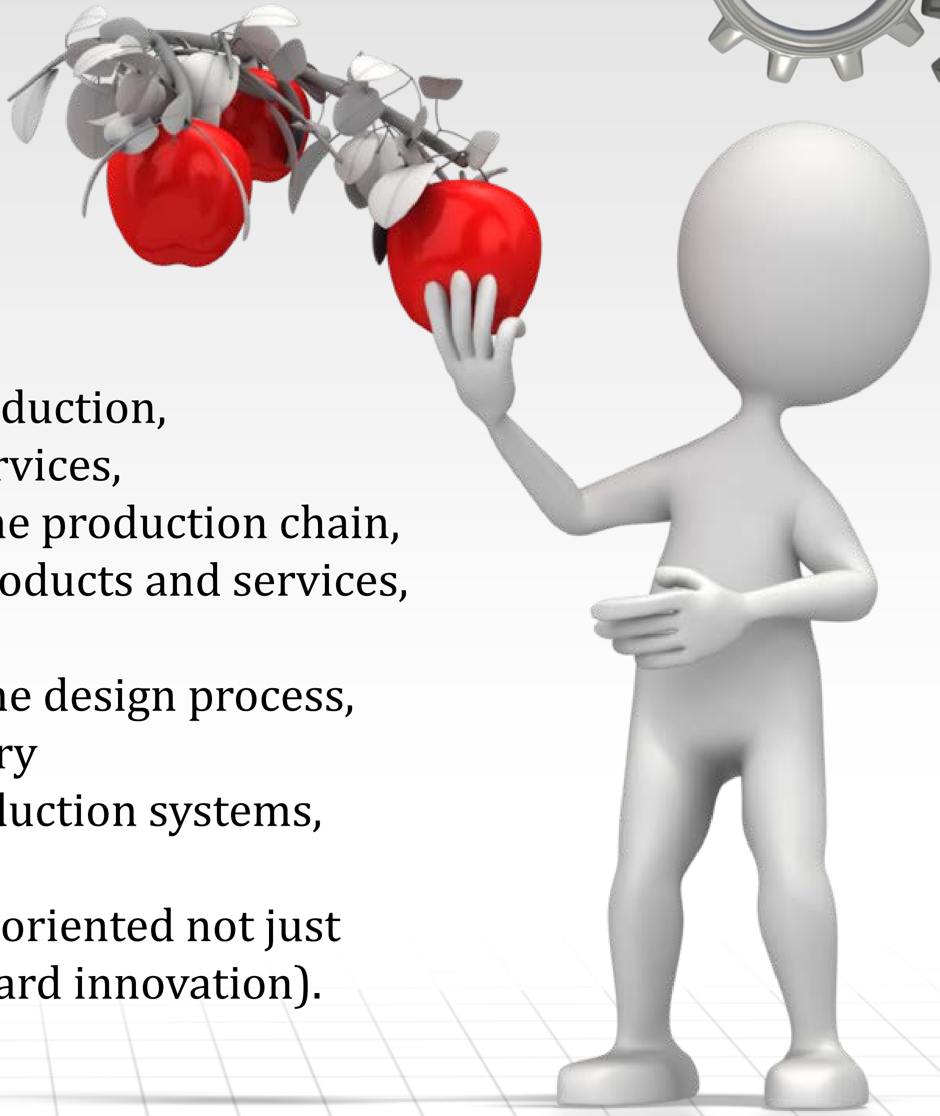
- Preconditions for implementing Industry 4.0:
  - standardization (systems, platforms, protocols, connections, interfaces,etc.),
  - process/work organization,
  - available products,
  - new bussines models,
  - security/know-how protection,
  - available skilled workers.

Industry 4.0 (4)

- Challenges in adopting an Industry 4.0 model:
  - data security issues,
  - high degree of reliability and stability,
  - maintaining the integrity pf the production process,
  - loss of high-paying human jobs,
  - avoiding technical problems.

- Required investments for:
  - integration of advanced ICT technologies,
  - purchasing modern equipment and advanced machine tools,
  - cutting costs with energy and materials,
  - integrating the clean tech practices,
  - training of staff and management.

Industry 4.0 (5)



- Benefits of Industry 4.0:
  - increasing flexibility,
  - increasing customization,
  - increasing the speed of production,
  - improving maintenance services,
  - streamlining activities in the production chain,
  - increasing the quality of products and services,
  - increased productivity,
  - customer involvement in the design process,
  - virtualization of the industry
  - interconnection of the production systems,
  - environmental compliance
  - change in business model (oriented not just toward profit, but also toward innovation).

Industry 4.0 in Romania

- Romania's advantages in the transition to industry 4.0:
  - bring back manufacturing to Europe with a focus on customized production, quality and close to the customer,
  - the industry 4.0 directly focuses on the automotive industry which in the past 10 years has grown a lot in Romania (mainly production of components)
  - internet speed connection,
  - the presence of many IT companies,
  - skills needed for digital manufacturing (numerous investments in R&D centers in the Automotive),
  - European community grants (2016-2020) for R&D in industry,
  - Germany, the main supporter of the Industry 4.0 is one of the largest investors in Romania.

Self-assessment test

1. Industry 4.0 uses cyber physical systems. T F

2. Industry 4.0 creates intelligent manufacturing systems. T F

3. Interoperability is high in Industry 4.0. T F

4. Romania is not a target for implementation of Industry 4.0. T F

5. Equipment used in factories that implement Industry 4.0 concepts have to be very reliable. T F

Industry 5.0

Industry 5.0 is defined by its use of advanced technologies to achieve societal goals alongside economic growth and business objectives.

- **Industry 5.0** will aim to:
  - take a human-centric approach towards manufacturing processes.
  - improve the sustainability of the manufacturing industry.
  - building resilience in the production process.
- The **differences** between Industry 4.0 and Industry 5.0:
  - The fourth revolution was driven by technological change, the fifth will be powered by values. New cyber systems will make manufacturing more sustainable, human-centric and resilient.
  - Industry 5.0 will prioritise greater collaboration between humans and machines through cyber-physical systems and technologies.

Conclusion

- Industry 4.0 is associated with the fourth industrial revolution, although some consider it just an evolution, not a revolution.
- The advanced technologies used initially for smartphones, appliances, cars, etc., have reached a certain maturity, becoming more interesting for industry.
- The trend of decentralization and globalization of production, makes suitable the integration of the IoT technologies and cyber-physical systems in production processes.
- However, there are significant concerns regarding the vulnerability of data in IoT systems, which can dramatically affect the integrity of production systems.

- Production systems - Review